

ANATOMY - MOCK MIDTERM 1

Head and neck: the face, the mouth and nose, and the eye

Instructions. 24 questions. Only **one answer** is correct for each question. Suggested time: 40 minutes. Answer on your own first, then check the answer key at the end, which explains each answer and names the trap the question uses. This paper was written for practice by Sina and is not an official exam.

Question 1/24 Which anatomical parts form the lateral wall of the orbit?

- A. The maxillary bone only
- B. The zygomatic bone and the greater wing of the sphenoid bone
- C. The zygomatic bone and the lesser wing of the sphenoid bone
- D. The lacrimal bone and the greater wing of the sphenoid bone

Question 2/24 The zygomatic bone:

- A. Contains a paranasal sinus
- B. Forms the medial wall of the orbit
- C. Articulates with the frontal, sphenoid, temporal and maxillary bones
- D. Gives insertion to the temporalis muscle

Question 3/24 Regarding the sphenoid bone:

- A. The foramen spinosum transmits the mandibular nerve
- B. The foramen rotundum is located in the lesser wing
- C. The sella turcica is located on the upper surface of its body
- D. The optic canal lies between the greater and the lesser wings

Question 4/24 The pterion:

- A. Is located at the back of the skull, near the lambda
- B. Is the thickest part of the lateral wall of the skull
- C. Is the meeting point of the frontal, parietal, squamous temporal bones and the greater wing of the sphenoid
- D. Overlies the posterior branch of the superficial temporal artery

Question 5/24 In the maxilla:

- A. The frontal process runs up between the nasal and lacrimal bones
- B. It is a cranial bone
- C. The alveolar process forms the floor of the orbit
- D. The infraorbital foramen transmits the mental nerve

Question 6/24 The bony nasal septum is formed by:

- A. Only the perpendicular plate of the ethmoid bone
- B. Mainly the vomer with smaller contribution from the palatine bone
- C. The vomer and the inferior nasal concha
- D. The perpendicular plate of the ethmoid bone and the vomer

Question 7/24 Which facial sinuses drain at the superior meatus?

- A. The sphenoid sinus
- B. The posterior ethmoidal cells
- C. The maxillary sinus
- D. The frontal sinus

Question 8/24 Regarding the maxillary sinus:

- A. It is supplied by the ophthalmic nerve
- B. Its floor is related to the roots of the upper molar teeth
- C. It is the smallest paranasal sinus
- D. Its ostium opens in the inferior meatus

Question 9/24 In the tongue:

- A. Protrusion is produced by the styloglossus muscle
- B. The hypoglossal nerve supplies all its muscles except the palatoglossus
- C. The posterior 1/3 carries the vallate papillae
- D. The foramen caecum is located on its ventral surface

Question 10/24 Regarding the salivary glands:

- A. The submandibular duct opens on the sublingual papilla, beside the frenulum
- B. The parotid gland receives its secretomotor fibers from the chorda tympani
- C. The sublingual gland is a purely serous gland
- D. The parotid duct opens opposite the lower second molar

Question 11/24 The muscles of facial expression:

- A. Are subcutaneous and insert into the skin
- B. Insert mainly into bone
- C. Include the masseter muscle
- D. Are supplied by the mandibular nerve

Question 12/24 Regarding the buccinator muscle:

- A. It arises in part from the pterygomandibular raphe
- B. It is a muscle of mastication
- C. It is pierced by the submandibular duct
- D. It is supplied by the buccal nerve of the mandibular nerve

Question 13/24 The lateral pterygoid muscle:

- A. Elevates the mandible
- B. Inserts on the angle of the mandible
- C. Pulls the condyle and the articular disc forward
- D. Is supplied by the facial nerve

Question 14/24 Temporo-mandibular joint:

- A. Its dislocation is usually posterior
- B. Its articular surfaces are covered by hyaline cartilage
- C. Is a fibrous joint
- D. Its articular disc divides it into two synovial cavities

Question 15/24 The following statement is correct:

- A. The lingual artery is a branch of the internal carotid artery
- B. The maxillary artery enters the skull through the foramen spinosum
- C. The occipital artery is a terminal branch of the external carotid artery
- D. The facial artery ends as the angular artery

Question 16/24 Regarding the veins of the face:

- A. The facial vein has no connection with the cavernous sinus
- B. The retromandibular vein is formed by the facial and lingual veins
- C. The external jugular vein drains into the internal jugular vein
- D. The angular vein is formed by the union of the supratrochlear and supraorbital veins

Question 17/24 Lymph from the lower lip and the chin drains first into:

- A. The preauricular nodes
- B. The occipital nodes
- C. The submandibular nodes
- D. The submental nodes

Question 18/24 Regarding the eyelid:

- A. The orbital septum connects the tarsi to the orbital margins
- B. The levator palpebrae superioris is supplied by the facial nerve
- C. The glands of Zeis are modified sweat glands
- D. The tarsal plates are made of hyaline cartilage

Question 19/24 The lacrimal gland:

- A. Is supplied by the facial artery
- B. Receives its secretomotor fibers through the pterygopalatine ganglion
- C. Opens into the inferior conjunctival fornix
- D. Is located in the inferomedial part of the orbit

Question 20/24 Aqueous humor is secreted into the posterior chamber and leaves the eye at the iridocorneal angle. The structure that secretes it is:

- A. The iris
- B. The choroid
- C. The lens
- D. The ciliary processes

Question 21/24 Regarding the retina:

- A. The optic disc contains the highest density of cones
- B. The fovea lies at the centre of the macula
- C. Bipolar cells generate action potentials
- D. Rods are responsible for colour vision

Question 22/24 The superior oblique muscle:

- A. Is supplied by the oculomotor nerve
- B. Elevates the eye
- C. Has a tendon that passes through the trochlea
- D. Arises from the anterior part of the floor of the orbit

Question 23/24 Obstruction of a vessel leads to damage of the left oculomotor nerve as it leaves the midbrain. On examination, the left eye is most likely:

- A. Turned inward only, with a normal pupil
- B. Turned up and in, with a small pupil
- C. Turned down and out, with a dilated pupil and ptosis
- D. Normal, with ptosis of the right eyelid

Question 24/24 When light is reflected on one eye, both pupils constrict due to the pupillary light reflex. The efferent limb of this reflex travels in:

- A. The oculomotor nerve and the ciliary ganglion
- B. The facial nerve
- C. The optic nerve
- D. The ophthalmic nerve

Answer key, mock midterm 1

Q	Ans	Why	Trap used
1	B	Lateral wall = zygomatic + greater wing of the sphenoid; it is the strongest wall.	Lesser/greater wing swap; 'only'.
2	C	It has four articulations and no sinus; temporalis inserts on the coronoid process.	Sinus: only the maxilla has one among facial bones.
3	C	Rotundum is in the greater wing; the superior orbital fissure (not the optic canal) lies between the wings; spinosum = middle meningeal artery.	Wing swap and foramen swap.
4	C	Thinnest point, over the anterior branch of the middle meningeal artery.	Thickest/thinnest swap; wrong artery.
5	A	Infraorbital foramen = infraorbital nerve; the orbital surface (not the alveolar process) forms the orbital floor; the maxilla is facial.	Nerve swap; process swap.
6	D	Bony septum = perpendicular plate of the ethmoid + vomer (septal cartilage completes it in front).	'Mainly... with smaller contribution', 'only' (same template as the hard palate question).
7	B	Superior meatus = posterior ethmoidal cells. Maxillary and frontal = middle meatus; sphenoid = spheno-ethmoidal recess.	Same question as the real mock, asked about a different meatus.
8	B	Largest sinus; floor = alveolar process; ostium in the middle meatus; V2.	Largest/smallest; wrong meatus; V1/V2.
9	B	Genioglossus protrudes; foramen caecum is at the apex of the sulcus terminalis on the dorsum; vallate papillae are just in front of the sulcus.	Muscle swap; dorsal/ventral; 1/3 vs 2/3.
10	A	Parotid duct: upper second molar; sublingual: mostly mucous; parotid secretomotor = IX via the otic ganglion.	Upper/lower; 'purely'; nerve swap.
11	A	Facial expression = VII, subcutaneous, into skin. Masseter is a muscle of mastication (V3).	VII/V3 swap.
12	A	It is a facial muscle (VII, buccal branch) pierced by the parotid duct.	Buccal nerve (V3, sensory) vs buccal branch of VII; duct swap.
13	C	It inserts on the neck, capsule and disc and starts opening of the mouth; V3.	The only muscle of mastication that does not close the jaw.
14	D	Synovial joint, fibrocartilage, dislocation usually anterior.	Same template as the real midterm question.
15	D	Lingual = ECA; the middle meningeal (a branch of the maxillary) uses the spinosum; terminal branches = superficial temporal + maxillary.	Real-paper template ('The following statement is correct').
16	D	Retromandibular = superficial temporal + maxillary; EJV drains into the subclavian; facial vein ↔ ophthalmic veins ↔ cavernous sinus.	Absolute 'no connection'.
17	D	Upper face → preauricular; cheek/upper lip → submandibular; lower lip/chin → submental.	Neighbouring node group.
18	A	Tarsi = dense connective tissue; Zeis = sebaceous (Moll = sweat); levator = CN III.	Zeis/Moll swap; III/VII swap.

Q	Ans	Why	Trap used
19	B	Superolateral, in the lacrimal fossa of the frontal bone; ducts into the superior fornix; lacrimal artery (ophthalmic).	Superior/inferior and lateral/medial swaps.
20	D	Ciliary processes secrete; trabecular meshwork / Schlemm's canal drain.	The stem gives the path but not the source.
21	B	Optic disc = blind spot; cones give colour; only ganglion cells fire action potentials.	Disc/fovea swap; rods/cones swap.
22	C	CN IV; from the body of the sphenoid; depresses, abducts, intorts. The inferior oblique arises from the floor.	Superior/inferior oblique swap.
23	C	Unopposed lateral rectus (VI) and superior oblique (IV); levator and sphincter pupillae lost.	Vignette in the style of the real Q19.
24	A	Afferent II → pretectal nuclei → both Edinger-Westphal nuclei → III → ciliary ganglion → sphincter pupillae.	Afferent/efferent confusion.